श्री गणेशाय :

Generalized Inverse (G-Inverse)

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned} \quad \Longleftrightarrow Ax = b$$

- Suppose A has a left inverse $\rightarrow A_L^{-1}A = I_n$

$Ax = b$

if $b \in C(A)$ then $x = A_L^{-1} b$

- Suppose A has right inverse.

$A_R^{-1} b$ $\qquad Ax = b \Rightarrow A(A_R^{-1} b) = b$

$\Rightarrow \underline{b = b}$.

$A_{m\times n}$ has a right inverse if $\exists\, B_{n\times m}$ s.t.

$AB = I_m \Longleftrightarrow \rho(A) = m$ ← rank(A)

- $A_{m\times n}$ has a left inverse if $\exists$ a matrix $B_{n\times m}$ st $BA = I_n$. $\rho(A) = n$

left inverse exists ⇒ full col' rank

& $A(A_L^{-1}b) = b$

* $Ax = b$

if $b \in C(A)$ then $A_L^{-1}b$ (or $A_R^{-1}b$) in a solution to $Ax = b$

- For a matrix $A_{m\times n}$, another matrix $G_{n\times m}$ is said to be a generalized inverse (g-inverse)

Defn

if $b \in C(A) \Rightarrow A(Gb) = b$

Theorem : Every matrix has a g-inverse.

Proof: $A_{m\times n} = 0$ the $0_{n\times m}$ is a g-inverse of $A_{m\times n}$

Suppose $A \neq 0$. Then $\rho(A) \geq 1$. Consider a RF of A say (P, Q)

$A = P_{m\times r} Q_{r\times n}$ & $\rho(P) = \rho(Q) = r$

* $G_1 = (Q_R^{-1} P_L^{-1})_{n\times m}$ is a g-inverse

Take a vector $b \in C(A) \Rightarrow b = Au_{n\times 1}$

$A(Gb) = A(Q_R^{-1} P_L^{-1})Ax = PQQ_R^{-1}P_L^{-1}PQx$

Fr

$= PP_L^{-1}PQx$

$= PQx = Ax = b$

- For a matrix $A_{m\times n}$, $Q_R^{-1} P_L^{-1}$ is a g-inverse of A. However, all the g-inverse of A is not of this form.

Theorem: Let $A_{m\times n}$ & $G_{n\times m}$. Then the following statements are equivalent.

(i) G is a g-inverse of A

(ii) $AGA = A$

(iii) AG is idempotent. $\rho(AG) = \rho(A)$

(iv) GA is idempotent $\rho(GA) = \rho(A)$.

Proof: (i) $\Rightarrow$ (ii)

$\forall x \in \mathbb{R}^n \quad Ab$ where $b = Ax$

$AGAx = Ax \Rightarrow (AGA - A)x = 0$

$\Rightarrow x \in N(AGA - A)$

Since it holds for all n, we have $N(AGA - A) = \mathbb{R}^n$

$\Rightarrow \rho(AGA - A) = 0 \Rightarrow AGA = A$

(ii) $\Rightarrow$ (iii) $AGA = A$

$\Rightarrow AGAG = AG$

$AGA = A$

$\Rightarrow \rho(AG) \geq \rho(A)$

$AB = C$ (Rank-Nullity Thm)

$\rho(C) \leq \rho(A)$

Also, $\rho(AG) \leq \rho(A)$. Thus $\rho(AG) = \rho(A)$

(iii) $\Rightarrow$ (iv) $C(AG) \subseteq C(A)$ $\quad C = AB$

$\Rightarrow C(AG) = C(A)$ $\quad C(C) \subseteq C(A)$

$\Rightarrow C(A) \subseteq C(AG)$

$\rightarrow A = AGX$ for some C

$AGAG = AG \Rightarrow AGAGL = AGC \Rightarrow AGA = A$

$\Rightarrow GAGA = GA$

(iv) $\Rightarrow$ (i)

$GAGA = GA$ and $\rho(GA) = \rho(A)$ (given)

We have to show

$b \in C(A) \Rightarrow A(Gb) = b$

$b = Az$ for some $z \in \mathbb{R}^r$

We know $\rho(GA) = \rho(A) \Rightarrow R(GA) = R(A)$

$\Rightarrow A = XGA$

Also,

$GAGA = GA \Rightarrow XGAGA = XGA \Rightarrow AGA = A$

$AGb = AGAz = Az = b$

* $Ax = b$ is consistent if $b \in C(A)$

* $Ax = 0$ the set of all possible solutions is $N(A)$.

- if $Ax = b$ is consistent then the set of all solutions is a particular sol^n + $N(A)$

Theorem: Let $A_{m \times n}$ be a matrix & G a g-inverse of A

(i) $Ax = b$ is consistent iff $AGb = b$

(ii) The solution space of $Ax = 0$ is $\{(I - GA)z : z \in \mathbb{R}^n\}$

(iii) if $Ax = b$ is consistent, then the set of all solutions in $\{Gb + (I - GA)z : z \in \mathbb{R}^n\}$

(i) ($\Rightarrow$) $Ax = b$ is consistent $\Rightarrow b \in C(A)$. Then by the def^n of g-inverse, $AGb = b$

($\Leftarrow$) $AGb = b \Rightarrow Ax = b \Rightarrow b \in C(A)$

(x = Gb)

We have to show

(ii) $N(A) = C(I - GA)$ Take $x \in N(A) \Rightarrow Ax = 0$

$\Rightarrow GAx = 0 \rightarrow (I - GA)x = x \Rightarrow x \in C(I - GA)$

$x \in C(I - GA) \Rightarrow x = (I - GA)y \Rightarrow x = y - GAy$

$\Rightarrow Ax = Ay - AGAy = Ay - Ay = 0$

$\Rightarrow$ So, $x \in N(A)$

(iii) Gb is a particular solution

Thus, the set of all solutions of $Ax = b$ is $\{Gb + (I - GA)z : z \in \mathbb{R}^n\}$

Suppose A has a left inverse A_L^{-1}

$A(A_L^{-1}b) = b \quad \forall b \in C(A)$

Let G be a g-inverse of A

Qn: Is G a left inverse of A?

$AGA = A \Rightarrow A_L^{-1}AGA = A_L^{-1}A = I$

$\Rightarrow GA = I$

So, G is a left inverse of A

- If A has left inverse $\Rightarrow$ the set of g-inverses of A is the same as the set of left inverse of A.

- If A has a right inverse $\Rightarrow$ the set of g-inverses of A is the same as the set of right inverses of A.

- If A has a inverse $\Rightarrow$ G inverse is unique & equals to A^{-1}

- Let G_1 & G_2 be two inverses of A & $G_1 \neq G_2$

$\alpha G_1 + (1-\alpha) G_2$ is also inverse of A $\quad 0 < \alpha < 1$

$$A(\alpha G_1 + (1-\alpha) G_2) A = \alpha A G_1 A + (1-\alpha) A G_2 A$$
$$= \alpha A + (1-\alpha) A$$
$$= A$$

$E_k E_{k-1} \cdots E_1 A = F$

$A_{m \times n}$ & $G_{n \times m}$ is a g-inverse of A

$$\rho(A) \leq \rho(G) \leq \min\{m, n\}$$

$\star$ $AGA = A$

$\rho(A) \leq \rho(AG) \leq \rho(G)$

$A = PQ$ is an RF of A

then $G = Q_R^{-1} P_L^{-1}$ is a g-inverse of A & $\rho(G) = \rho(A)$ (Verify!)

$\therefore$ all G inverse are not of this form. So, we present below thm.

Theorem: For any s such that $\rho(A) \leq s \leq \min\{m, n\}$ there is a g-inverse of A say G, with $\rho(G) = s$

Proof :- $E = \begin{bmatrix} X_{r \times r} & 0 \\ 0 & 0 \end{bmatrix}_{m \times n}$ s.t. X^{-1} exists.

Then $G = \begin{bmatrix} X^{-1} & B \\ C & D \end{bmatrix}_{n \times m}$ is a g-inverse of E.

$$EGE = \begin{bmatrix} X & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} X^{-1} & B \\ C & D \end{bmatrix} \begin{bmatrix} X & 0 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} I_r & XB \\ 0 & 0 \end{bmatrix} \begin{bmatrix} X & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} X & 0 \\ 0 & 0 \end{bmatrix} = E$$

Suppose $A_{m\times n}$ is a matrix with $\rho(A) = r$

The $A = P\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q$ where $P_{m\times m}$ & $Q_{n\times n}$ are non-singular matrices. - - - - (1)

$A_{m\times n} = \bar{P}_{m\times r}\ \bar{Q}_{r\times n}$ where $\rho(A) = r$

Construct $P = [\bar{P}\ \hat{P}]_{m\times m}$ where the columns of $\hat{P}$ & $\bar{P}$ form a basis of $\mathbb{R}^m$. This means P is a non-singular matrix.

Similary, $Q = \begin{bmatrix} \bar{Q} \\ \hat{Q} \end{bmatrix}$ where rows of $\bar{Q}$ & $\hat{Q}$ forms a basis of $\mathbb{R}^n$. Thus, Q is n-s matrix.

$$P\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q = [\bar{P}_{m\times r}\ \hat{P}_{m\times(n-r)}] \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \bar{Q} \\ \hat{Q} \end{bmatrix}$$

$$= [\bar{P}\ 0] \begin{bmatrix} \bar{Q} \\ Q \end{bmatrix} = \bar{P}\bar{Q} = A$$

Suppose $A_{m\times n}$ is a matrix with $\rho(A) = r$

Then

$A = P\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} Q$ where P & Q are n-s matrices.

We claim for any matrix PMQ where P & Q are n-s matrices, $Q^{-1} M^{-1} P^{-1}$ is a g-inverse of PMQ.

For A, A^{-1} is a g-inverse.

$PMQQ^{-1}M^{-}P^{-}PMQ = PMM^{-}MQ = PMQ$

$A = P\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}Q$

$Q^{-1}\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}^{-}P^{-1}$ is a g-inverse of A

$\Rightarrow Q^{-1}\begin{bmatrix} I_r & B \\ C & D \end{bmatrix}P^{-1}$ is a g-inverse of A

Take $B = C = 0$ & D has $s-r$ rows linearly independent.

Result: $\rho\left(\begin{bmatrix} A & B \\ 0 & D \end{bmatrix}\right) \geq \rho(A) + \rho(D)$ (Verify)

Using this, we can show that $\rho\left(\begin{bmatrix} I_r & B \\ C & D \end{bmatrix}\right) \geq \rho(I_r) + \rho(D)$

$= r + s - r = s$. Further, because there are only s non-zero rows, which implies

$$\rho\left(\begin{bmatrix} I_n & B \\ C & 0 \end{bmatrix}\right) \leq s$$

Computation of a g-inverse

$E_{m\times m}A_{m\times n} = F_{m\times n}$ where F is in a reduced echelon form.

Construct $G_{n\times m}$ where p_1^{th} row is the first row of E, p_2^{th} row in the second row of F, ... p_r^{th} row in the r^{th} row of E, & all the other row are 0.

We claim G is a g-inverse of A.

Proof: We will show that $\rho(GA) = \rho(A)$ & GA is idempotent.

p_i^{th} row of GA will be i^{th} row of F. $\forall\ 1 \le i \le r$ & all other row are 0. This together with $\rho(F) = \rho(A)$ implies

$$\rho(GA) = \rho(A)$$

AB

i-th row of AB is $(A_i)B$

i^{th} row from A $\times$ Full col. from B.

$$(AB)_{i\cdot} = [(AB)_{i1}\ (AB)_{i2} \cdots (AB)_{in}]$$

$$= (A_i)B$$

GA will have r non-null rows there are p_1^{th} row, p_2^{th} row, ... p_r^{th} row & all the other rows will be zero.

$$(AB)_{i\cdot} = (A)_{i\cdot} B$$

$$(AB)_{\cdot j} = A(B)_{\cdot j}$$

1st - $(p_1 - 1)^{th}$ column : 0

p_1^{th} column : e_{p_1}

$(p_1+1)^{th}$ column to $(p_2 - 1)^{th}$ column

: $\alpha_j e_{p_i}$

$(p_i + 1 \le j \le p_2 - 1)$

p_2^{th} column : e_{p_2}

$(p_2+1)^{th}$ - $(p_3 - 1)^{th}$ column : $\beta_j e_{p_1} + \gamma_j e_{p_2}$

$p_2 + 1 \le j \le p_3 - 1$

p_r^{th} column - e_{p_r}

$(p_r + 1)^{th}$ column - n^{th} column.

$\delta_1 e_{p_1} + \delta_2 e_{p_2} + \cdots + \delta_r e_{p_n}$

$p_r + 1 \le j \le n$

$$(GA)(GA) = |GA|$$

We will show that $(GA)(GA)_{*j} = (GA)_{*j}$

$\forall\ 1 \le j \le r$

This holds for first $(p_1 - 1)$ columns

$$(GA)(GA)_{p_1} = (GA)e_{p_1} = (GA)_{p_1}$$

$$(GA)(GA)_j = \alpha_j (GA)e_{p_1} = \alpha_j (GA)_{p_1}$$

$$= \alpha_j e_{p_1}$$

$$= (GA)_j$$

$p_1 + 1 \le j \le p_2 - 1$

$$(GA)(GA)_j = \beta_j (GA)e_{p_1} + \gamma_j (GA)e_{p_2}$$

$$= (GA)_j$$

Hence $(GA)(GA) = GA$.

Lemma:

Let $A_{m \times n}$ be a matrix. Then $A^T(AA^T)^-$ and $(A^TA)^-A^T$ are g-inverses of A.

Proof: $AA^T(AA^T)^-AA^T = AA^T$

$$\rho(AA^T) = \rho(A) = \rho(A^TA)$$

$$e(AA^T) = e(A)$$

$$A = AA^TC$$

$\rho(AA^T)$

$= \rho(A^T) - d(\mathcal{C}(A^T) \cap \mathcal{N}(A))$

$= \rho(A^T) = \rho(A)$

$$AA^T(AA^T)^-AA^TC = AA^TC$$

$$\Rightarrow A[A^T(AA^T)^-]A = A$$

$[(AA^T)(A^T)^-]$

Some properties of G

1) Reflexivity : $\rho(G) = \rho(A) \iff GAG = G$

2) Minimum norm : $b \in C(A) \to \|Gb\| \leq \|x\|$,

$\forall x$ s.t. $\forall x$ s.t. $Ax = b$

$\iff (GA)^T = GA$

3) Least square : $\forall x \in \mathbb{R}^n$, $\forall y \in \mathbb{R}^m$

$\|AGy - y\| \leq \|Ax - y\|$

$\iff (AG)^T = AG$

1) $(\Rightarrow)$ $\rho(A) = \rho(AG) \leq \rho(G) \Rightarrow \rho(G) = \rho(AG)$

$\Rightarrow G = CAG$

$AGA = A$

$GAG = C(AGA)G = C(A)G = G$

So, $\boxed{GAG = G}$

$(\Leftarrow)$ $GAG = G \Rightarrow \rho(G) \leq \rho(AG) = \rho(A)$

$\rho(G) \leq \rho(A)$

$\rho(G) \geq \rho(A)$ [Basic proposition]

$\therefore \boxed{\rho(A) = \rho(G)}$

2.) $(\Leftarrow)$ $b \in C(A) \Rightarrow Ax = b \iff x = Gb + (I - GA)z$

$z \in \mathbb{R}^n$

$\|x\|^2 = \|Gb\|^2 + \|(I - GA)z\|^2 + 2b^T G^T (I - GA)z$

$b = Aw$

$2b^T G^T (I - GA)z = 2w^T A^T G^T (I - GA)z$

$= 2w^T (GA - (GA)^2)z$

$= 2w^T (GA - GA)z = 0$

$$\therefore \quad \|Ax\| \geq \|GAb\| \qquad \forall\, Ab$$

$$\Rightarrow \quad \|GAb\|^2 + \|(I-GA)z\|^2 + 2b^T G^T (I-GA)z$$

$$\geq \|(I-GA)z\|^2 + 2b^T G^T (I-GA)z \geq 0$$

$$\text{"} \quad \text{"} \quad + 2\omega^T A^T G^T (I-GA)z \geq 0$$

Replace ω with $\alpha\omega$. (as if ω is a soln then $\alpha\omega$ is also a soln)

$2\alpha\omega^T A^T G^T (I-GA)z$

If $\omega^T A^T G^T (I-GA)z$ is negative we can choose α large enough s.t. the expression < 0

If $\omega^T A^T G^T (I-GA)z$ is positive we can choose α small enough s.t. expression < 0

$$\therefore \quad \omega^T A^T G^T (I-GA)z \qquad \forall\, \omega, \forall z$$

$$A^T G^T (I-GA) = 0$$

$$\left((GA)^T\right)^T = \left((GA)^T GA\right)^T \quad \text{--- (1)}$$

$$GA = (GA)^T GA \quad \text{--- (2)} \qquad \text{How ?} \qquad \therefore \quad \boxed{GA = (GA)^T}$$

Using (1) & (2)

($\Leftarrow$)

$$\|Ax-y\|^2 = \|AGy - y + A(x-Gy)\| = \|AGy-y\|^2$$

$$+ \|Aw\|^2$$
$$+ 2w^T A^T (AGy - y)$$

$$2w^T A^T (AG-I)y = 2w^T (A^T AG - A^T)y$$
$$= 2w^T \left((AGA)^T - A^T\right)y$$

$$= 0$$

$$\therefore \quad \boxed{\|Ax-y\| \geq \|AGy-y\|}$$

Least square g-inverse

$$\|AGy - y\| \leq \|Ax - y\| \quad \forall x, y$$

$$\Leftrightarrow (AG)^T = AG$$

Given $\|AGy - y\| \leq \|Ax - y\| \quad \forall x, y$

$$\Rightarrow \|AGy - y\| \leq \|AGy - y + A\omega\| \quad \omega = x - Gy \quad \forall \omega, y$$

$$\Rightarrow \|AGy - y\|^2 \leq \|AGy - y\|^2 + \|A\omega\|^2 + 2\omega^T A^T (AG - I)y$$

$$\|A\omega\|^2 + 2\omega^T A^T (AG - I)y \geq 0 \quad \forall \omega, y$$

$$\Leftrightarrow \omega^T A^T (AG - I)y = 0 \quad \forall \omega, y$$

$$\Leftrightarrow A^T (AG - I) = 0$$

$$\Leftrightarrow A^T AG = A^T$$

$$\Leftrightarrow G^T A^T A G = G^T A^T = (AG)^T$$

$$\Rightarrow AG = (AG)^T$$

Moore - Penrose g-inverse

A g-inverse of a matrix A is a Moore-Penrose g-inverse if it is a (i) reflexive, (ii) minimum norm, and (iii) least-square g-inverse

$\Leftrightarrow G$ is a MP g-inverse of A if

(i) $AGA = A$ (ii) $GAG = G$ (iii) $(GA)^T = GA$

(iv) $(AG)^T = AG$

Thm every matrix has a MP g-inverse

Proof - If $A_{m\times n}=0$ then $G_{n\times m}=0$ is the MP g-inverse

suppose A is non-null. consider an RF of A say (P,Q)

let $G = Q^T (P^T A Q^T)^{-1} P^T$

$$(P^T A Q^T)^{-1} = (P^T P Q Q^T)^{-1} = (Q Q^T)^{-1} (P^T P)^{-1}$$

We claim A is a MP g-inverse

$$AGA = A Q^T (Q Q^T)^{-1} (P^T P)^{-1} P^T A$$

$$= P (Q Q^T) (Q Q^T)^{-1} (P^T P^{-1}) (P^T P) Q$$

$$= PQ = A$$

$$GAG = Q^T (P^T A Q^T)^{-1} P^T A Q^T (P^T A Q^T)^{-1} P^T$$

$$= Q^T (P^T A Q^T)^{-1} P^T$$

$$= G$$

Suppose A is non-null. Consider an RF of A, say (P,Q) Let $G = Q^T (P^T A Q^T)^{-1} P^T$

Thm: MP g-inverse is unique

$\rho(G) \le \rho(Q^T) = r$ &
$\rho(G) \ge \rho(A) = r$
Hence $\rho(G) = r$

Take two MP g-inverses, say G_1 & G_2

$G_1 = G_1 A G_1$ → $G_1 G_1^T A^T G_2^T A^T$

$G_1 G_1^T A^T$

$= G_1 A G_1 G_2^T A^T$

$= G_1 A G_1$

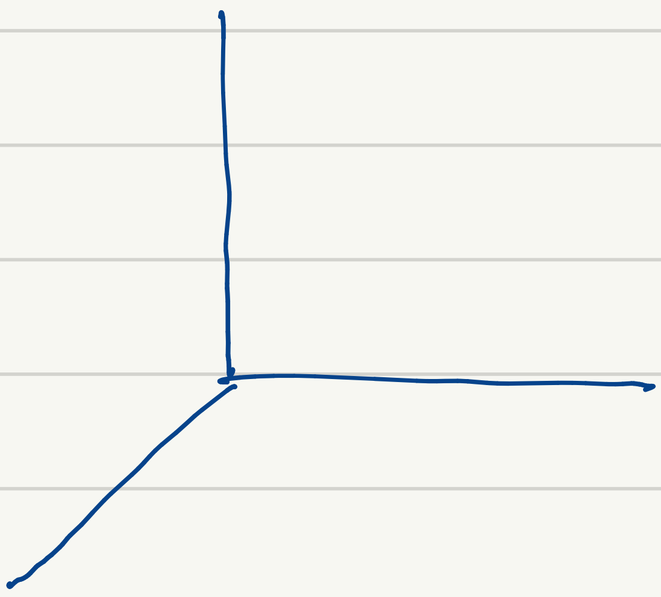
$$G_2 = G_2 A G_2 = A^T G_2^T G_2 = A^T G_1^T A^T G_2^T G_2$$

$$= A^T G_1^T G_2 A G_2 = G_1 A G_2$$

Projection

S & T are two subspace of $\mathbb{R}^n$

$sp(S \cup T) = S + T = \{u \in \mathbb{R}^n \mid u = x + y$ where $x \in S$ & $y \in T\}$



S = X-Y plane

T = Y-Z plane

$sp(S \cup T) = \mathbb{R}^3$

$u = (u_1, u_2, u_3) = (u_1, u_2, 0) + (0, 0, u_3)$

$\in S \qquad \in T$

$= (u_1, 0, 0) + (0, u_2, u_3)$

$\in S \qquad \in T$

Direct sum

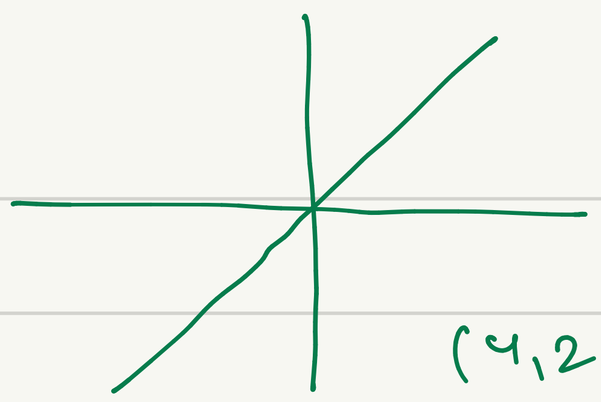
$S \oplus T = \{u \in \mathbb{R}^n \mid u = x + y$ where $x \in S$ & $y \in T$ & x & y are unique$\}$

A subspace of $\mathbb{R}^n$ is a complement of S (a subspace of $\mathbb{R}^n$)

if $S \oplus T = \mathbb{R}^n$ (They aren't unique)

Defn Let T be a complement of a subspace S & $u \in \mathbb{R}^n$. Then projection of u into S along T is x where $u = x + y$ for some

$x \in S$ & $y \in T$



S = x-axis

T = y = x line

(4,2) = (2,0) + (2,2)

T' = y = 2x

i.e.

Theorem: Let T be a complement of S & P(u) denote the projection of u into S along T. Then,

(i) the projection of u into T along S is $u - p(u)$

(ii) $p(u) = u$ iff $u \in S$

(iii) $p(u) = 0$ iff $u \in T$

(iv) $p(p(u)) = p(u)$

(v) $p(u+v) = p(u) + p(v)$

(vi) $p(\alpha u) = \alpha p(u)$

Proof: (i) $u = p(u) + y$

$\Rightarrow y = u - p(u)$

(ii) $u \in S$ $\quad u = u + 0 \Rightarrow p(u) = u$

$p(u) = u$ $\quad u = p(u) + y \Rightarrow y = 0$

$u = u + 0 \in S$

(iv) $p(u) \in S$ $\quad p(p(u)) = p(u)$

(v) $u = x + y$, $v = w + z$, $u + v = (x+w) + (y+z)$

$p(u) = x$, $p(v) = w$ $\quad p(u+v) = x + w$

(vii) $u \in \mathbb{R}^n$ $\quad u = x + y$ $\quad \alpha u = \alpha x + \alpha y$

$\in S \quad \in T$

$S, T \subseteq \mathbb{R}^n$

S & T are complements of each other

We will try to get a matrix $P_{n \times n}$ s.t. for any vector $u \in \mathbb{R}^n$ Pu is the projection of u into S along T.

Defn A matrix $P_{n \times n}$ is called a projector into S along T if for $u \in \mathbb{R}^n$, Pu is the projection of u into S along T.

P is a projection matrix if P is a projector into S along T for some S & T where T is a complement of S.

$Py = 0$

$P^2 y = Px = 0$

$Py = Px = 0$

$P \cdot (I-P)z = 0$

$C(P) \cap N(P) = \{0\}$

Take $x \in T$

Thm: The following are eqn..

(i) P is a projection matrix

(ii) $P^2 = P$

(iii) $N(P) = C(I-P)$

(iv) $\rho(P) + \rho(I-P) = n$

Proof: (i) $\Rightarrow$ (ii) Assume P projects into S along T. Take $x \in \mathbb{R}^n$

then Px is the projection of S into T.

$P(Px) = Px \Rightarrow P^2 x = Px \Rightarrow (P^2 - P)x = 0$

$x \in N(P^2 - P) \quad \forall x \in \mathbb{R}^n \Rightarrow P^2 - P = \mathbb{R}^n$

Hence $\rho(P^2 - P) = 0 \Rightarrow \boxed{P^2 = P}$.

(ii) $\Rightarrow$ (iii)

$x \in N(P) \Rightarrow Px = 0 \Rightarrow x - Px = x \Rightarrow (I-P)x = x$

$\Rightarrow x \in C(I-P) \Rightarrow x = (I-P)z$ for some $z \in \mathbb{R}^n$

$\Rightarrow \quad Px = P(I-P)z = (P-P^2)z = 0 \Rightarrow x \in N(P)$

Now, $\quad x \in C(I-P) \quad \therefore \quad x = (I-P)z$ for $z \in \mathbb{R}^n$

$\Rightarrow \quad Px = P(I-P)z = (P-P^2)z = 0 \Rightarrow$
$x \in N(P)$

(iii) $\Rightarrow$ (iv) $\quad d(N(P)) = d(C(I-P))$

$\Rightarrow \quad n - \rho(P) = \rho(I-P)$

$\rho(P) + \rho(I-P) = r$

(iv) $\Rightarrow$ (i) $\quad I-P = Q$

$\Rightarrow \quad P+Q = I$

and by (iv) $\quad \rho(P) + \rho(Q) = \rho(I)$

Thus $C(Q)$ is a complement of $C(Q)$

$x \in \mathbb{R}^n, \quad x = Ix = (P+Q)x = Px+Qx$

So, $C(Q)$ is a complement of $C(P)$ and Px is the projection of vector x into $C(P)$ along $C(Q)$

Hence P is a projection matrix.

Note:- If P is a projection matrix, then P projects into $C(P)$ along $C(I-P)$ & $(I-P)$ projects into $C(I-P)$ along $C(P)$

Qn. suppose T is a complement of S & P is a projector into S along T. Is P unique?

$Pe_1 = P_{\cdot 1}$, $Pe_2 = P_{\cdot 2}$, ...

Therefore P is unique.

Existence of a projection matrix

Projection matrix always exists

construction of a projection matrix

Let T be a complement of S. Take a basis of S & a basis of T & make matrix B &C. Construct the following matrix $A = [B \mid C]$ where $e(B) = S$ & $e(C) = T$

clearly $\rho(A) = n$. write $A^{-1} = \begin{bmatrix} G \\ H \end{bmatrix}$ s.t. BG is defined.

Claim BG is a projector into S along T

Proof: $AA^- = I \Rightarrow [B \mid C] \begin{bmatrix} G \\ H \end{bmatrix} = I$

$\Rightarrow BGB + CHI = I$

$x \in \mathbb{R}^n \Rightarrow x = \underset{\in S}{BGx} + \underset{\in T}{CHx}$

Also, $e(BG) \subseteq e(B)$ and $e(CH) \subseteq e(C)$

$\Rightarrow e(BG) \cap e(CH) = \{0\}$

$\rho(BG) + \rho(I - BG) = n \Rightarrow BG$ is a projection matrix and it projects into $e(BG)$ along $e(I - BG)$

$$\ell(BG) \subseteq \ell(B)$$

$$d(\ell(BG)) \leq r \quad - (1)$$

$$\rho(BG) \geq \rho(B) + \rho(G) - r = R + R - R = R \quad - (2)$$

from (1) & (2)

$$\Rightarrow \star \; \rho(BG) = r \Rightarrow \ell(BG) = \ell(B)$$

$\star$ S is a subspace of $\mathbb{R}^n$ & $x \in \mathbb{R}^n$

$$\min_{w \in S} \|x - w\|$$

Defⁿ A projector P into S along T is called an orthogonal projector if T is the orthogonal complement of S.

S, $S^{\perp} = \{x \in \mathbb{R}^r \mid \langle x, y \rangle = 0 \;\; \forall y \in S\}$

Theorem :- Let S be a subspace of $x \in \mathbb{R}^n$
Then $\arg\min_{w \in S} \|x - w\|$ = orthogonal projection x into S

$$\|x - w\|^2 = \|x - Op(x) + Op(x) - w\|^2$$

$$= \|x - Op(x)\|^2 + \|Op(x) - w\|^2 + 2(Op(x) - w)^T (x - Op(x))$$

$x - Op(x) \in S^{\perp}$ & $Op(x), w \in S$

$Op(x) - w \in S$

$(Op(x) - w)^T (x - Op(x)) = 0$

$$\Rightarrow \quad \|x-w\|^2 = \|x-Px\|^2 + \|Px-w\|^2$$

$$\geq \|x-Px\|^2$$

$$\therefore \quad w = Px$$

Orthogonal projection matrix

P is orthogonal projection matrix $\Longleftrightarrow$ P is idempotent & symmetric

$$P = P^T$$

$$= P \cdot P$$

$$P = P^T P$$

$\therefore P = P^2$

P projects into $C(P)$ along $C(I-P)$

To show $C(P) \perp C(I-P)$

Take vectors Px & $(I-P)y$

$$\langle Px, (I-P)y \rangle = x^T P^T (I-P) y$$

$$= x^T (P^T - P^T P) y$$

$$= x^T (P - P^2) y$$

$$= 0$$

$\therefore C(P) \perp C(I-P)$

Converse.

Suppose P is orthogonal projection matrix $P^2 = P$ & $C(P) \perp C(I-P)$

$$x^T P^T (I-P) y = 0$$

$$x^T (P^T - P^T P) y = 0$$

$\forall x, y$

$$\therefore \quad P^T = P^T P \quad - (1)$$

$$P = P^T P \quad - (2)$$

$$\therefore P = P^T$$

Computation of orthogonal projection matrix

Consider $A_{m \times n}$ s.t. $\rho(A) = S$

$$P_A = A (A^T A)^- A^T$$

is a projection matrix into S

Need to show:

(i) $P_A = P_A^T P_A$

(ii) $\rho(P_A) = \rho(A)$

(i) $A (A^T A)^- A^T = A \underbrace{(A^T A)^- A^T}_{\text{g-inverse of } A} A (A^T A)^- A^T$

$$= A (A^T A)^- A^T$$

(ii) $\rho(P_A) \subseteq \rho(A)$

$P_A A = A$

$\rho(A) \subseteq \rho(P_A)$

$\therefore \quad \rho(P_A) = \rho(A)$

$\therefore$ P_A is orthogonal projection matrix into S.

$\rho(A) = \rho(B) \Rightarrow P_A = P_B$

$$e(A) = e(D) \Rightarrow P_A = P_B$$

D is matrix with orthonormal col.

$$P_D = D (D^T D)^{-} D^T$$

$$= D D^T \qquad [D^T D = I_r]$$

$$S_1 + S_2 + \dots + S_k$$

$$S_1 + S_2 + S_3 = sp(S_1 \cup S_2 \cup S_3)$$

PI: $sp(S_1 \cup S_2 \cup S_3) \subseteq S_1 + S_2 + S_3$

↳ show that this is a subspace

$$S_1 \subseteq S_1 + S_2 + S_3$$

$$\Downarrow$$

$$\cup S_i \subseteq S_1 + S_2 + S_3$$

$$\Downarrow$$

$$sp(\cup S_i) \subseteq S_1 + S_2 + S_3$$

$$S_1 + S_2 + S_3 \subseteq sp(S_1 \cup S_2 \cup S_3)$$

$$x \in S_1 + S_2 + S_3$$

$$x = x_1 + x_2 + x_3$$

$$\in S_1 \quad \in S_2 \quad \in S_3$$

$$x_1 \in sp(S_1 \cup S_2 \cup S_3)$$

$$x_2 \in sp(S_1 \cup S_2 \cup S_3)$$

$$x_3 \in sp(S_1 \cup S_2 \cup S_3)$$

$$\therefore \quad x \in sp(S_1 \cup S_2 \cup S_3)$$

$$\therefore \quad \boxed{S_1 + S_2 + S_3 = sp(\cup S_i)}$$

$S_1 \oplus S_2 \oplus S_3$ ~~$\Leftrightarrow$~~ $S_i \cap S_j = \{0\}$, $\forall\, i, j \in \{1,2,3\}$

$\Downarrow$

$\{(1,0,0)\}$, $\{(0,1,0)\}$, $\{(1,1,0),(0,0,1)\}$

$\Leftrightarrow$ $S_1 \cap S_2 = \{0\}$ & $(S_1+S_2)\cap S_3 = \{0\}$

Thm: Equivalent

1. $S_1 + S_2 + S_3$ is direct
2. $S_1 \cap S_2 = \{0\}$ and $(S_1+S_2)\cap S_3 = \{0\}$
3. $d(S_1+S_2+S_3) = d(S_1)+d(S_2)+d(S_3)$

Proof: (1) $\Rightarrow$ (2)

$x \in S_1 \cap S_2$

$\Rightarrow$ $x \in S_1+S_2+S_3$

$x = x+0+0 = 0+x+0$

$\in S_1$ $\in S_2$

$\therefore$ decomposition is unique

$x = 0$

$\therefore$ $S_1 \cap S_2 = \{0\}$

$x \in (S_1+S_2)\cap S_3$

$x = x_1+x_2+0 = 0+0+x$

$\Downarrow$

$x = 0$

(ii) $\rightarrow$ (i)

Assume sum isn't direct

$x \in S_1+S_2+S_3$

$x = y_1+y_2+y_3 = z_1+z_2+z_3$

$y_i \neq z_i$

$$(y_3 - z_3) = (z_1 - y_1) + (z_2 - y_2)$$

$\in S_3$ $\quad \in S_1$ $\quad \in S_2$

But $S_3 \cap (S_1 + S_2) = \{0\}$

(contradiction)

$\therefore$ Sum is direct

(ii) $\Rightarrow$ (iii)

$$d(S+T) = d(S) + d(T) - d(S \cap T)$$

$$d(S_1 + S_2 + S_3) = d(S_1 + S_2) + d(S_3)$$

$$d(S_1 + S_2 + S_3) = d(S_1) + d(S_2) + d(S_3)$$

(iii) $\Rightarrow$ (ii)

$$d(S_1 + S_2 + S_3) \leq d(S_1 + S_2) + d(S_3)$$

$\Downarrow$

$(S_1 + S_2) \cap S_3 = \{0\}$

$d(S_1 + S_2) \leq d(S_1) + d(S_2)$

$\Downarrow$

$S_1 \cap S_2 = \{0\}$

(Projection (from Rao - Bhimasankaran)
Sect - 1.7, 3.7, 7.5)

$$S_1 + S_2 + S_3 = sp(S_1 \cup S_2 \cup S_3)$$

$S_1 \oplus S_2 \oplus S_3 = \mathbb{R}^n$ $\qquad$ $S \oplus T = \mathbb{R}^n$

Suppose $S_1 \oplus S_2 \oplus S_3 = \mathbb{R}^n$ & $x = x_1 + x_2 + x_3$ where $x_i \in S_i \ \forall\, i \in \{1,2,3\}$ then x_i is called the projection of x into S_i along S_2 & S_3. Similarly we can define projection into S_2 & S_3.

Let P_i be the projector into S_i along $\{S_j \mid j \neq i\}$

$$P_1 + P_2 + P_3 = I_n$$

For any vector $x \in \mathbb{R}^n$, we have $(P_1 + P_2 + P_3)x = x$

$$\Rightarrow (P_1 + P_2 + P_3)x = x$$

$$\Rightarrow (P_1 + P_2 + P_3 - I_n)x = 0 \quad \forall x \in \mathbb{R}^n$$

$$\Rightarrow N(P_1 + P_2 + P_3) = \mathbb{R}^n$$

$$\Rightarrow P_1 + P_2 + P_3 = I_n$$

$S \oplus T = \mathbb{R}^n$ P is the projection into S along T

$(I - P)$ is the projection along S.

Fischer Cochran Theorem

Let $A_1, A_2 \ldots A_k$ be matrices of order n s.t. $\sum_{i=1}^{k} A_i = I_n$. Then, the following statements are equivalent

(i) each A_i is a projection matrix

(ii) $\sum_{i=1}^{k} \rho(A_i) = n$

(iii) $A_i A_j = 0 \quad \forall \ i \neq j$

Proof:

(i) $\rightarrow$ (ii) each A_i is an idempotent matrix

$$\rho(A_i) = tr(A_i) \quad \forall \ i$$

We have $\sum_{i=1}^{k} A_i = I_n$

$$\Rightarrow \quad tr\left(\sum_{i=1}^{k} A_i\right) = n$$

$$\Rightarrow \quad \sum_{i=1}^{k} tr(A_i) = n$$

$$\Rightarrow \quad \sum_{i=1}^{k} \rho(A_i) = n$$

(ii) $\Rightarrow$ (iii)

Let (P_i, Q_i) be RF of A_i $\forall i$

$A_i = P_i Q_i$ & $\rho(A_i)$ = no. of colums of P_i

$$P = \underset{n\times n}{[P_1 \; P_2 \ldots P_k]} \qquad Q = \begin{bmatrix} Q_1 \\ Q_2 \\ \vdots \\ Q_k \end{bmatrix}_{n\times n}$$

As $\sum \rho(A_i) = n$, we have P is $n\times n$ & Q is $n\times n$.

$$PQ = [P_1 \ldots P_k] \begin{bmatrix} Q_1 \\ \vdots \\ Q_k \end{bmatrix}$$

$$= P_1Q_1 + P_2Q_2 + \ldots P_kQ_k$$

$$= A_1 + A_2 + \ldots + A_k = I_n$$

As P has n many columns & Q has n many rows, we have (P,Q) is an RF of I_n.

$A^2 = A \Leftrightarrow QP = I_r$ where $r = \rho(A)$

$$QP = I_n$$

$$\begin{bmatrix} Q_1 \\ Q_2 \\ \vdots \\ Q_K \end{bmatrix} \begin{bmatrix} P_1 \cdots P_K \end{bmatrix} = I_n$$

$$\Rightarrow \begin{bmatrix} Q_1P_1 & Q_1P_2 \cdots \\ Q_2P_1 & Q_2P_2 \cdots \\ \vdots & \vdots \\ & \cdots Q_KP_K \end{bmatrix} = \begin{bmatrix} I_{r_1} & O \cdots & O \\ O & I_{r_2} & \vdots \\ \vdots & & I_{r_K} \end{bmatrix}$$

where r_i = rank of A_i

$$Q_i P_j = O \quad \forall\, i \neq j$$

$$\Rightarrow P_i Q_i P_j Q_j = O \quad \forall\, i \neq j$$

$$\Rightarrow A_i A_j = O \quad \forall\, i \neq j$$

(iii) $\Rightarrow$ (i)

$$A_i (A_1 + A_2 + \cdots + A_K) = A_i$$

$$\Rightarrow A_i A_i = A_i \Rightarrow A_i^2 = A_i$$

Partial derivative (Ref. Harville chap 15)

$f : \mathbb{R}^n \to \mathbb{R}$

if $n=2$, $f : \mathbb{R}^2 \to \mathbb{R}$, $\left.\frac{\partial f(x,y)}{\partial x}\right|_{(x,y)=(a,b)}$

$$= \lim_{h \to 0} \frac{f(a+h, b) - f(a,b)}{h}$$

Ily we can define $\left.\frac{\partial f}{\partial y}\right|_{(x,y)=(a,b)}$

We can define partial derivatives for a function $f: \mathbb{R}^n \to \mathbb{R}$ and $\frac{\partial f}{\partial x} = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \cdots \frac{\partial f}{\partial x_n}\right]^T_{x=(x_1,\ldots x_n)}$

$\frac{\partial^2 f}{\partial x_i^2}$ is the partial derivative of the function $\frac{\partial f}{\partial x_i}$ wrt x_i

$\frac{\partial^2 f}{\partial x_i \partial x_j}$ is the partial derivative of the function

$\frac{\partial f}{\partial x_j}$ w.r.t. x_j

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

To check whether f has a local maxima or minima at y, compute H at y. Then

(i) if $H|_y$ is a pd matrix then H has a local minima at y.
↳ positive definite

(ii) if $H|_y$ is a nd matrix then H has a local maxima at y.
↳ negative definite

pd matrix (Positive definite)

A matrix $B_{n \times n}$ called a positive matrix if $x^T B x > 0 \quad \forall x \in \mathbb{R}^n / \{0\}$

Negative definite

A matrix $B_{n \times n}$ is called negative definite matrix if $x^T B x < 0 \quad \forall x \in \mathbb{R}^n / \{0\}$

$f(x) = a^T x \quad \forall x \in \mathbb{R}^n$

$$[a_1 \; a_2 \; \cdots \; a_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\frac{\partial f}{\partial x} = [a_1, a_2, \ldots a_n]^T = a^T$$

$f(x) = x^T A x \quad \forall x \in \mathbb{R}^n$

$$= \begin{bmatrix} 2a_{11}x_1 + (a_{12}+a_{21})x_2 + \cdots + (a_{1n}+a_{n1})x_n \\ \vdots \\ (a_{n1}+a_{1n})x_1 + \cdots + 2a_{nn}x_n \end{bmatrix}$$

$$= (A + A^T)x$$

$$H = \begin{bmatrix} 2a_{11} & a_{12}+a_{21} & \cdots & a_{1n}+a_{n1} \\ \vdots & & & \\ a_{n1}+a_{1n} & \cdots & a_{n2}+a_{2n} & \cdots 2a_{nn} \end{bmatrix}$$

$$= A + A^T$$

$f: \mathbb{R}^{m \times n} \to \mathbb{R}$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f}{\partial x_{11}} & \frac{\partial f}{\partial x_{12}} & \cdots & \frac{\partial f}{\partial x_{1m}} \\ \frac{\partial f}{\partial x_{21}} & \cdots & \cdots & \frac{\partial f}{\partial x_{2n}} \\ \frac{\partial f}{\partial x_{n1}} & \frac{\partial f}{\partial x_{n2}} & \cdots & \frac{\partial f}{\partial x_{nn}} \end{bmatrix}$$

$f(X) = \operatorname{tr}[A_{n\times m} X_{m\times n}]$ where $A_{n\times m}$ is fixed

$$\frac{\partial f}{\partial x_{ij}} = \frac{\partial}{\partial x_{ij}}[\operatorname{tr}(AX)] \xrightarrow{\text{one method}} = \frac{\partial}{\partial x_{ij}}\left(\sum_{i=1}^{n}\sum_{k=1}^{m} a_{ik}x_{ki}\right)$$

↳ $\operatorname{tr}\left[A \frac{\partial X}{\partial x_{ij}}\right]$

how? ↳ $= \operatorname{tr}[A e_i e_j^T] = \operatorname{tr}[e_j^T A e_i]$

$= a_{ji}$

$$\frac{\partial X}{\partial x_{ij}} = \begin{bmatrix} 0 & \cdots & \overset{j^{th}\text{ col.}}{\downarrow} & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{bmatrix} \leftarrow j^{th}\text{ row} = \underset{m\times 1}{e_i} \underset{1\times n}{e_j^T}$$

$$\frac{\partial tr(AX)}{\partial x_{ij}} = a_{ji} \qquad \frac{\partial X}{\partial x_{ij}} = \underset{n\times 1}{e_i}\,\underset{1\times n}{e_j^T}$$

$$\Rightarrow \quad \frac{\partial tr(AX)}{\partial X} = A^T$$

$$f : \mathbb{R}^{m\times n} \longrightarrow \mathbb{R}$$

$$\frac{\partial F}{\partial X} = \begin{bmatrix} \frac{\partial f}{\partial x_{11}} & \cdots & \frac{\partial f}{\partial x_{1n}} \\ \vdots & & \\ \frac{\partial f}{\partial x_{m1}} & \cdots & \frac{\partial f}{\partial x_{mn}} \end{bmatrix}$$

$$\frac{\partial X}{\partial x_{ij}} = \begin{bmatrix} \frac{\partial x_{11}}{\partial x_{ij}} & \cdots & \frac{\partial x_{1n}}{\partial x_{ii}} \\ \vdots & & \\ \frac{\partial x_{m1}}{\partial x_{ij}} & \cdots & \frac{\partial x_{mn}}{\partial x_{ij}} \end{bmatrix}$$

If X is symmetric

$$\frac{\partial X}{\partial x_{ii}} = e_i e_i^T \qquad \frac{\partial X}{\partial x_{ij}} = e_i e_j^T + e_j e_i^T$$

$$\frac{\partial tr(AX)}{\partial x_{ii}} = a_{ii} \qquad \frac{\partial tr(AX)}{\partial x_{ij}} = tr\left(A \frac{\partial X}{\partial x_{ij}}\right)$$

$$= tr\left(A\, e_i e_j^T + e_j e_i^T\right)$$

$$= tr\left(A e_i e_j^T + A e_j e_i^T\right)$$

$$= tr\left(A e_i e_j^T\right) + tr\left(A e_j e_i^T\right)$$

$$= tr\left(\underbrace{e_j^T A}_{(A_{j*})_{1\times n}} e_i\right) + tr\left(e_i^T A e_j\right)$$

$$= a_{ji} + a_{ij}$$

$$\frac{\partial tr(AX)}{\partial X} = \begin{bmatrix} a_{11} & a_{12}+a_{21} & \cdots & a_{1n}+a_{n1} \\ \vdots & & & \\ a_{n1}+a_{1n} & \cdots & \cdots & a_{nn} \end{bmatrix}$$

$$= A + A^T - \text{diag}(a_{11}, \ldots a_{nn})$$

$$\frac{\partial \, tr(X)}{\partial X} = I_n$$ (In both cases, if X is symmetric or not)

$$\det(X) = \sum_{j=1}^{n} x_{ij} X_{ij} \quad \forall i \in \{1, \ldots n\}$$

where X_{ij} is the co-factor x_{ij}.

Differentiation of determinant of X

X is a square matrix of order n

$$\frac{\partial \det(X)}{\partial X} \qquad \frac{\partial \det(X)}{\partial x_{ij}}$$

$$\frac{\partial \det(X)}{\partial x_{ij}} = \sum_{k=1}^{n} X_{ik} \frac{\partial x_{ik}}{\partial x_{ij}} = X_{ij}$$

$$\frac{\partial \det(X)}{\partial X} = \begin{bmatrix} X_{11} & X_{12} & \cdots & X_{1n} \\ X_{21} & & & \vdots \\ \vdots & & & \vdots \\ X_{n1} & \cdots & \cdots & X_{nn} \end{bmatrix} = (adj(X))^T$$

Differentiation

Linear models

$y_1, y_2, \ldots y_n$ are the random variables, which are the outcomes of the experiment

Basic assumptions

(i) $E(Y_i) = x_{i1}\beta_1 + x_{i2}\beta_2 + \ldots + x_{ip}\beta_p \quad \forall i \in [n]$

where βs are unknown and $x_{i1}, \ldots x_{in}$ are known

$E(Y) = X\beta$ where $Y = [Y_1, Y_2, \ldots Y_n]^T$ and

$$X = \begin{bmatrix} x_{11} & x_{12} & \ldots & x_{1p} \\ x_{21} & x_{22} & \ldots & x_{2p} \\ \vdots & & & \vdots \\ x_{n1} & x_{n2} & \ldots & x_{np} \end{bmatrix}$$

(ii) $\text{Var}(y_i) = \sigma_{ii} = \sigma^2 (\text{unknown}) \quad \forall i \in [n]$

$\text{cov}(y_i, y_j) = 0 \quad \forall i \neq j$

$(x_1 \ldots x_n) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = [x_1, x_2 \ldots x_n]^T$

Alternatively → x is known & β is unknown

$y = x\beta + \varepsilon$ where $\varepsilon = (\varepsilon_1, \ldots \varepsilon_n)$ is a random vector

$E(\varepsilon_i) = 0$ & $\text{Var}(\varepsilon_i) = \sigma^2 \quad \forall i \in [n]$

$\text{cov}(\varepsilon_i, \varepsilon_j) = 0 \quad \forall i \neq j$

(Check whether this model is same or not)

Goal: To estimate $\beta_1, \ldots \beta_p$ and their linear functions Also, to estimate σ^2

A linear function of β say $\ell^T\beta$, is estimable if $\exists$ a linear function $c^T y$ s.t. $E(c^T y) = \ell^T\beta$

Theorem:

A linear function $\ell^T\beta$ is estimable iff $\ell^T \in R(X)$

Proof: As $\ell^T\beta$ is estimable, $\exists$ a linear function $c^T y$

s.t. $E(c^T y) = \ell^T\beta$

$\Rightarrow c^T E(y) = \ell^T\beta \Rightarrow c^T X\beta = \ell^T\beta \Rightarrow (c^T X - \ell^T)\beta = 0$

$\Rightarrow (c^T X - \ell^T) = 0$ ($\because$ it holds for all β)

$\Rightarrow \ell^T = c^T X \Rightarrow \ell^T \in R(X)$

Suppose $\ell^T \in R(X)$

$\Rightarrow \ell^T = c^T X$

$\Rightarrow \ell^T\beta = c^T X\beta = c^T E(y) = E(c^T y)$

This means $\ell^T\beta$ is estimable

Suppose $\rho(X_{n\times p}) < p$

$(1,0,0\ldots0)$, $(0,1,0\ldots)$ $\ldots$ $(0,0,\ldots,1)$ will not be all in $R(X)$

Thm: Suppose $\ell^T\beta$ is estimable & G is a LS g-inverse of X (least square)

Then $\ell^T G y$ is an estimator of $\ell^T\beta$ & among all the linear estimators of $\ell^T\beta$. $\ell^T G y$ has minimum variance.

Proof: $E(\ell^T G y) = \ell^T G E(y) = \ell^T G X\beta$

Now as $\ell^T\beta$ is estimable we have $\ell^T = c^T X$

$(XG)^T = XG$

$E(\ell^T G y) = \ell^T G X\beta = c^T X G X\beta = c^T X\beta = \ell^T\beta$

Take a linear estimator of $\ell^T\beta$, say $a^T y$

Claim: $a^T y = (\ell^T G + w^T) y$ with $w^T X = 0$

Proof: $a^T y = a^T y + \ell^T G y - \ell^T G y = \ell^T G y + (a^T - \ell^T G) y$

w^T

$w^T X = (a^T - \ell^T G) X$

$= a^T X - \ell^T G X$

$= a^T X - c^T X G X = a^T X - c^T X$

$X_1, X_2, \ldots X_n$ is a random sample from $F(\theta)$ where θ is unknown.

- Goal is to estimate θ or $h(\theta)$, a function of θ

- we estimate θ using a function of the random sample, $T(x_1, \ldots x_n)$

- $T(X_1, \ldots X_n)$ is called an estimator of θ
- we look for an estimator T s.t.

(i) $E(T) = \theta$ (an unbiased estimator of θ)

(ii) $V(T) \leq V(S)$ where S is another estimator with $E(S) = \theta$

and G a LS g-inverse of X

Theorem Let $l^T\beta$ be estimable. Then l^TGy is an unbiased estimator $l^T\beta$ and it has the minimum variance among all linear unbiased estimator of $l^T\beta$ Further $var(l^TGy) = l^T(X^TX)^-l$

l^TGy is called a best linear unbiased estimator (BLUE) $l^T\beta$

Proof: $E(l^TGy) = l^TGE(y) = l^TGX\beta = u^TXGX\beta$

l^TGy is an unbiased estimator of $l^T\beta$

Take c^Ty, a linear unbiased estimator of $l^T\beta$

$E(c^Ty) = l^T\beta \Rightarrow c^TX\beta = l^T\beta \Rightarrow c^TX = l^T$

$$c^Ty = c^Ty + l^TGy - l^TGy = l^TGy + (c^T - l^TG)y$$
$$= l^TGy + w^Ty$$

$$w^TX = (c^T - l^TG)X = (c^TX - l^TGX)$$

$$= c^TX - u^TXGX = c^TX - u^TX$$
$$= c^TX - l^T = 0$$

Claim: $Var(l^TGy) < Var(l^TGy + w^Ty)$ where $w^TX = 0$

$$Var(l^TGy + w^Ty) = \sigma^2(l^TG + w^T)(l^TG + w^T)^T$$

$$= \sigma^2(l^TG + w^T)(G^Tl + w)$$

$$= \sigma^2(l^TGG^Tl + l^TGw + w^TG^Tl + w^Tw)$$

$Var(a^Ty) = a^TVar(y)a$

$l^T G w = u^T X G w = u^T G^T X^T w = 0$

$w^T G^T l = w^T G^T X^T u = w^T X G u = 0$

$var(l^T G y + w^T X') = \sigma^2 (l^T G G^T l + w^T w)$

$\geq \sigma^2 (l^T G G l)$

$var(y) = \sigma^2 I_n$

$var(y_i) = \sigma^2$

$cov(y_i, y_j) = 0$

$var\, y = \sigma^2 I_n$

$Var(l^T G y) = Var((G^T l)^T y) = \sigma^2 (G^T l)^T (G^T l)$

$= \sigma^2 (l^T G G^T l)$

$Var(l^T G y + w^T y) \geq Var(l^T G y)$

$l^T G y$ is a BLUE.

we have to show

$\sigma^2 (l^T G G^T l) = \sigma^2 (l^T (X^T X)^- l)$

$X^T X \; (G G^T) \; X^T X = X^T X G X^T X = X^T X$

$l^T (X^T X)^- l = u^T X (X^T X)^- X^T u$

Result: AC^-B suppose $R(A) \subseteq R(C)$ & $C(B) \subseteq C(C)$

$AC^-B = XCC^-CY = XCY$

- $R(X) \subseteq R(X^T X)$ & $C(X^T) \subseteq C(X^T X)$

$\sigma^2 (u^T X (X^T X) X^T u)$ is independent of the choice of the g-inverse of $X^T X$.

$l^T \in R(X)$
$\Rightarrow l^T = u^T X$

Thus $\sigma^2(\ell^T G G^T \ell) = \sigma^2(\ell^T (X^T X)^- \ell)$

$Y = X\beta + \mathcal{E}$ where X is known, β is unknown

$E(\mathcal{E}) = 0$ $\quad V(\mathcal{E}) = \sigma^2 I_n$ $\quad \sigma^2$ is unknown

- $\ell^T\beta$ is estimable iff $\ell^T \in R(X)$
- $\ell^T G y$, where G is L.S. g-inverse of X, is the best linear unbiased estimator of $\ell^T\beta$.

and $V(\ell^T G y) = \sigma^2 \ell^T (X^T X)^- \ell$

Now consider X s.t. $\rho(X) = r$, i.e. of full row rank (Gauss Markov set up)

$R(X) = R^p$

Every linear function of β is estimable

$$\beta = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} \qquad \beta_i = e_i^T \beta$$

The BLUE of β is $\hat{\beta} = \begin{bmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} = \begin{bmatrix} e_1^T G y \\ e_2^T G y \\ \vdots \\ e_p^T G y \end{bmatrix}$

where G is a LS g-inverse of X

$= I_p G y = G y$

$$XGX = X \Rightarrow \quad X^T G^T X^T = X^T \Rightarrow X^T X G = X^T$$

$$\Rightarrow G = (X^T X)^- X^T$$

$$y_{n\times 1} = X_{n\times p}\, \beta_{p\times 1} + \epsilon_{n\times 1}$$

$$P(X) = R^p$$

$X_{n\times p}$

$(X^T X)_{p\times p}$

$$\hat{\beta} = Gy = (X^T X)^{-1} X^T y$$

$$Var(\hat{\beta}) = Var(Gy) = \sigma^2 G G^T$$

$$= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1}$$

$$= \sigma^2 (X^T X)^{-1}$$

$Var(Ay) = A\, Var(y) A^T$

non-negative a number

$Var(y_1)$

$Var(l^T y)$ a non-negative no. $= l^T Var(y) l$

$$Var(y) = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & & & \vdots \\ \vdots & & & \\ \sigma_{n1} & - & - & \sigma_{nn} \end{bmatrix}$$

where $Var(y_i) = \sigma_{ii}$

$Cov(y_i, y_j) = \sigma_{ij}$

$Var(Ay) = A\, Var(y) A^T$

$$= \begin{bmatrix} \sigma^2 & 0 & 0 & 0 \\ 0 & \sigma^2 & & 0 \\ \vdots & & \ddots & \\ 0 & 0 & 0 & \sigma^2 \end{bmatrix}$$

$$\widehat{l^T \beta} = l^T \hat{\beta}$$

$$Var(l^T \hat{\beta}) = l^T Var(\hat{\beta})\, l$$

$$= \sigma^2\, l^T (X^T X)^- l$$

$$y = X\beta + \epsilon \quad \text{where} \quad E(\epsilon) = 0 \quad Var(\epsilon) = \sigma^2 I_n$$

we will get an estimator of β by minimising $\epsilon^T\epsilon$

$$\epsilon^T\epsilon = (y - X\beta)^T(y - X\beta)$$

Let G be a LS g-inverse of X. Then $\beta = Gy$ minimizes $(y - X\beta)^T(y - X\beta)$

$(X^TX)^- X^T$ is a LS g-inverse of X (verify)

$$\hat{\beta} = Gy = (X^TX)^- X^T y$$

$$(y - X\beta)^T(y - X\beta) = (y^Ty - 2y^TX\beta + \beta^TX^TX\beta)$$

$$\frac{\partial \epsilon^T\epsilon}{\partial \beta} = -2(y^TX)^T + (X^TX + X^TX)\beta$$

$$\& \quad (X^TX)\beta = X^Ty$$

- normal equation

$$\frac{\partial a^Tx}{\partial x} = a$$

$$\frac{\partial x^TAx}{\partial x} = (A + A^T)x$$

$$X^Ty \in \mathcal{C}(X^T) = \mathcal{C}(X^TX)$$

$(X^TX)\beta = X^Ty$ is consistent

$\&$ $\beta = (X^TX)^- X^Ty$ for any g-inverse of X^TX.

$H = X^TX$ is a p.d. matrix

Thus $\hat{\beta} = (X^TX)^- X^Ty$ minimises $\epsilon^T\epsilon$

$$E(\hat{\beta}) \overset{?}{=} \beta \qquad E((X^TX)^- X^Ty) = (X^TX)^- X^TE(y)$$

$$= (X^TX)^- X^TX\beta \neq \beta$$

$$(y - x\hat{\beta})^T (y - X\hat{\beta}) = (y - X(X^TX)^- X^Ty)^T$$
$$(y - X(X^TX)^- X^Ty)$$

$$= (y - P_X y)^T (y - P_X y)$$
$$= y^Ty - y^T P_X y - y^T P_X y + y^T P_X y$$
$$= y^T (I_n - P_X) y$$

Theorem: $\rho(X) = r$ then $\frac{1}{n-r} (y - X\hat{\beta})^T (y - X\hat{\beta})$ is an unbiased estimator of σ^2

Proof: $(y - X\hat{\beta})^T (y - X\hat{\beta}) = y^T (I_n - P_X) y$

$$E(y^T (I_n - P_X) y) = E((X\beta + \varepsilon)^T (I_n - P_X)(X\beta + \varepsilon))$$
$$= E(\beta^T X^T (I_n - P_X) X\beta + \varepsilon^T (I_n - P_X) X\beta + \beta^T X^T (I_n - P_X)\varepsilon + \varepsilon^T (I_n - P_X)\varepsilon)$$

$$= E(\varepsilon^T \underbrace{(I_n - P_X)}_{Q} \varepsilon) = E(\sum_{i,j} a_{ij} \varepsilon_i \varepsilon_j)$$
$$= \sum_{i,j} a_{ij} E(\varepsilon_i \cdot \varepsilon_j)$$
$$= \sum_{i=1}^{n} \omega_{ii} \sigma^2 = \sigma^2 tr(Q)$$
$$= \sigma^2 tr(I_n - P_X)$$
$$= \sigma^2 (n - r)$$

$$E\left(\frac{1}{n-r} y^T (I_n - P_X) y\right) = \sigma^2$$

- Is BLUE unique?
- Does there exist an unbiased estimator of β_i always?
- when does $l^T\beta$ have a unique unbiased linear estimator?

Ex: $y_{ij} = \alpha_i + \beta_j + \epsilon_{ij}$ $\quad i = 1,2 \quad j = 1,2$

$$E = \begin{bmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

$E(\epsilon) = 0$ & $Var(\epsilon) = \sigma^2 I_r$ where σ^2 is unknown

$l^T\gamma$ estimable $\iff l^T \in R(X) =$

$\{(x_1, x_2, x_3, x_4) \mid x_1 + x_2 = x_3 + x_4\}$

Suppose we want to get BLUE of $\alpha_1 + \beta_1$

$\alpha_1 + \beta_1 = l^T\gamma = (1,0,1,0)\gamma$

$l^T G y = l^T (X^T X)^- X^T y$

$$X^T X = \begin{bmatrix} 2 & 0 & 1 & 1 \\ 0 & 2 & 1 & 1 \\ 1 & 1 & 2 & 0 \\ 1 & 1 & 0 & 2 \end{bmatrix} \qquad (X^T X)^- = \frac{1}{4} \begin{bmatrix} 0 & 4 & -2 & -2 \\ 0 & -2 & 3 & 1 \\ 0 & -2 & 1 & 3 \end{bmatrix}$$

$l^T\gamma = u^T X \gamma$ where $u^T = (1,0,0,0)$

$u^T X (X^T X)^- X^T y$

$$X(X^TX)^-X^T = \frac{1}{4}\begin{bmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{bmatrix}$$

$$y^TX(X^TX)^-X^Ty = \frac{1}{4}(3y_{11} + y_{12} + y_{21} - y_{22})$$

$$V(y^TX(X^TX)^-X^Ty) = \frac{9}{16}\sigma^2 + \frac{1}{16}\sigma^2 + \frac{1}{16}\sigma^2 + \frac{1}{16}\sigma^2 + \frac{1}{16}\sigma^2$$

$$= \frac{3}{4}\sigma^2$$

Example :-

$$y_{ij} = \alpha_i + \epsilon_{ij} \qquad \forall\, i = 1, 2 \ldots, n$$

$$j = 1, 2, \ldots n_j$$

$$X = \begin{matrix} n_1 \\ \\ \\ \\ \\ \\ \end{matrix}\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

$E(\epsilon) = 0$

$V(\epsilon) = \sigma^2 I_{(n_1 + \ldots n_r)}$

$n_1 = 2, \quad n_2 = 1, \quad n_3 = 3$

$$X = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

$\mu + \alpha_1, \quad \mu + \alpha_2, \quad \mu + \alpha_3$

$\mu, \quad \alpha_1 - \alpha_2, \quad \alpha_1 \qquad l^T\beta = \mu$

$$[1, 0, 0, 0]\begin{bmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \mu$$

$[0, 1, 1, 0]$

Find BLUE of $\mu+\alpha_1, \mu+\alpha_2, \mu+\alpha_3$